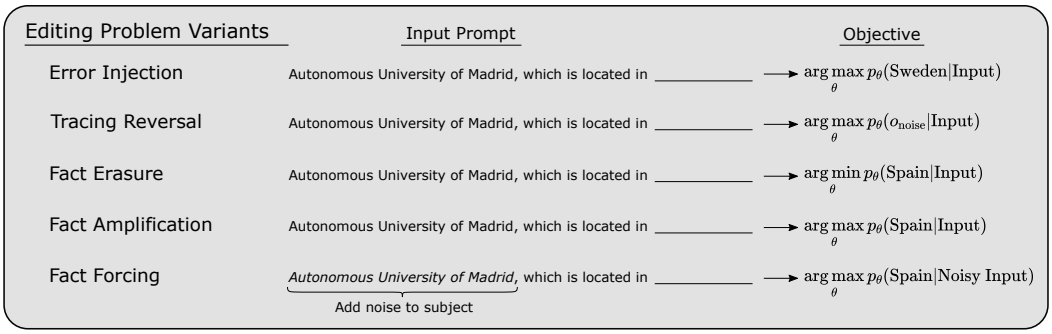


Figure 5: Depiction of editing problem variants. Rather than inject a new false fact into a model (Error Injection), we consider injecting the output obtained from noising the subject entity (Tracing Reversal), erasing a stored fact (Fact Erasure), amplifying a stored fact (Fact Amplification), or forcing a known fact onto the same kind of noisy input as used in Causal Tracing (Fact Forcing).

Additional Robustness Experiments. We include additional results in Appendix B using another dataset, ZSRE [20] (Figs. 19 and 20, Table 8), and another localization method, representation zeroing [2] (Figs. 21 and 22). Further robustness experiments in Appendix C include results with (1) other measures of edit success including Paraphrase Score, Neighborhood Score, and an Overall Score (Tables 4, 5 and 6), (2) different values of the tracing window size (Fig. 12), (3) GPT2-XL rather than GPT-J (Fig. 13), (4) the original unscaled metrics from Meng et al. [21] (Fig. 14), and (5) tracing effects measured at the last subject token rather than the max across tokens (Fig. 16). We find that **all of these experiments corroborate our results comparing Causal Tracing to Rewrite Score for GPT-J on CounterFact.** Considering these robustness results alongside additional editing method experiments that we consider in Sec. 5 below, we note that our main conclusions generalize across different causal localization methods, editing methods, editing metrics, models, and datasets.

5 Reconciling Localization and Editing

If injecting a new fact has little to do with where an existing fact is stored in the model, perhaps there is some other editing intervention that would be more closely related to insights from tracing analysis. In this section, we propose a few variants of the model editing problem that appear more and more like Causal Tracing in terms of their input, target, and objective. Then, we repeat and extend our analysis from Sec. 4 for all of these editing problems.

5.1 Editing Problem Variants

We summarize the following editing problems in Fig. 5.

1. *Error Injection*. The editing problem considered in Sec. 4, the objective being to maximize $p_\theta(o_{false}|s,r)$.
2. *Tracing Reversal*. We maximize $p_\theta(o_{noise}|s,r)$, aiming to change the model output from o_{true} back to the output for the "original" noised input $P = (s_{noise}, r)$ in Causal Tracing, o_{noise}.
3. *Fact Erasure*. Knowing where a fact is stored could be more useful for erasing the fact rather than injecting a new one. Hence, we consider erasing a fact by minimizing $p_\theta(o_{true}|s,r)$.
4. *Fact Amplification*. We reinforce known facts in the model by maximizing $p_\theta(o_{true}|s,r)$. Even for correctly predicted points, this value is often not near 1, leaving room for it to be increased.
5. *Fact Forcing*. As in Causal Tracing, this method uses a noised subject representation s_{noise}. We force the model to output o_{true} for this input by maximizing $p_\theta(o_{true}|s_{noise},r)$. Though this problem is of little practical significance, it is the most similar to Causal Tracing in its design, since it uses the same input as Causal Tracing and matches the goal of increasing the probability of o_{true} (see Sec. 3.2).

Note that solutions to each of these problems are evaluated according to our Rewrite Score, Paraphrase Score, and Neighborhood Score metrics from Sec. 3.4. The only difference is in the target output for the rewrite and paraphrase metrics (neighborhood is entirely identical).

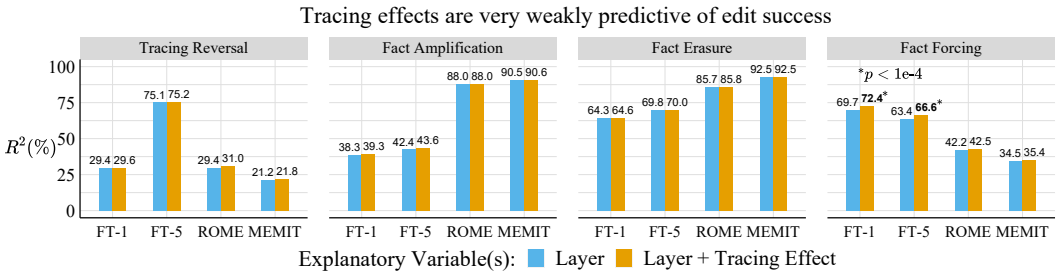


Figure 6: Tracing effects are very weakly predictive of edit success across editing problems and methods. Relative to the R^2 of a regression predicting rewrite score based on the edit layer (blue), a regression with edit layer and tracing effects (orange) improves the R^2 by at most .03 points (bolded). The choice of edit layer is a much better predictor of the rewrite score.

5.2 Experiment Design and Additional Edit Methods

We use the same experimental procedure as in Sec. 4, except that we consider a broader set of editing methods besides ROME. We list the four methods below:

1. ROME. The edit method from Sec. 4, ROME edits a single MLP layer's down-projection weight.
2. MEMIT. Though designed to edit multiple facts at once, when editing a single fact this method differs from ROME only by spreading out its update over several layers rather than one layer [22].
3. Constrained Finetuning (window size 1). We adopt a simple Adam-based optimization approach with an ℓ_∞-norm constraint, following Zhu et al. [42]. The window size of 1 indicates we apply this method at a single layer.
4. Constrained Finetuning (window size 5). The above finetuning method on five adjacent layers.

We select these methods for their simplicity and since ROME and MEMIT are designed specifically to edit MLP layers. Note that we report results for Causal Tracing with a window size of five, so when we use MEMIT or constrained finetuning to edit five layers, these five layers can exactly match the range of restored layers from Causal Tracing.

5.3 Experiment Results

Main Results. As in our analysis in Sec. 4, we report R^2 values for a linear regression model predicting the rewrite score based on (1) the choice of edit layer treated as a categorical variable, or (2) that variable interacted with the tracing effect. We show the results in Fig. 6, with R^2 values for each regression above their respective bars (numbers also in Appendix Table 3). We find that, relative to the Layer-only regression, **tracing effects explain at most an additional 3.2% of the variance in edit success** across our different editing problems and editing methods. This is a very small effect, especially compared to R^2 values from the Layer-only regression, which explains most of the variance in the outcome (58.5% on average across conditions in Fig. 6). We believe this is surprising given how the editing problem variants are designed. **It would seem that knowing where a fact is stored should help with amplifying or erasing that fact, but our results appear to fully disconfirm this hypothesis.** Interestingly, it also appears that it makes little difference whether we edit at one layer or five layers in order to match the number of representations restored by Causal Tracing. Based on comparisons between finetuning methods (FT-1 and FT-5) and between ROME and MEMIT (applied to 5 layers), editing at five layers does not improve the alignment between tracing and editing. In addition to our robustness results listed in Sec. 4.3, we also repeat our analysis using a subset of points where tracing effects are concentrated to a small number of layers, in order to focus on points where MEMIT and FT-5 edit *all* of the layers where the fact is stored. Results are nearly identical for this subset of the data (see Appendix B).

One Successful Case. We see the strongest positive relationship between edit success and tracing effects for Fact Forcing with finetuning methods. Here, we find that tracing effects explain an additional 3% of the variance in edit success (up from 1.5% for other experiments). This effect

is statistically significant at $p < 1e{-}4$ according to an F-test[2] comparing the two models (see visualization in Appendix Fig. 17). The result for Fact Forcing suggests that using s_{noise} rather than s in the model input is the cause of the positive relationship between editing and localization. We rule out the choice of target and maximizing vs. minimizing the target probability as possible causes based on the design of each problem variant (see Fig. 5): (1) the choice of target is not important since results are similar for Error Injection, Tracing Reversal, and Fact Amplification, and (2) maximizing vs. minimizing the target probability is not important since results are similar for Fact Erasure and Fact Amplification. Yet, tracing effects are still weakly informative of Fact Forcing editing if they explain only 3% of the variance in edit success. This points to there being other deeper reasons for localization results being unrelated to editing success.

6 Discussion

Does Causal Tracing tell us anything? We show that Causal Tracing is not indicative of which layer to select for model editing. However, this does not mean that localization insights from Causal Tracing have been useless. Causal Tracing has helped reveal the role that early-to-mid-range MLP representations *at the last subject token index* play in factual association in autoregressive language models, and ROME does perform better on average when optimizing the last subject token representation rather than another token representation [21].[3] Past work finds that both MLP and attention layers can show large Causal Tracing effects, and additional empirical editing experiments then demonstrate that it is preferable to edit MLP weights [21].

Why is edit success high at layers where the edited fact is not actually stored? First, we note that information is gradually accumulated across layers in a Transformer forward pass, as discovered by past work [31, 12, 21, 22, 13]. We suggest that it is possible to "override" the information in layer ℓ with an edit to another layer k (where $k < \ell$ or $k > \ell$). Since ROME is typically effective across a large range of layers (see Fig. 9), it appears that ROME can override the information accrued across 5 or 10 layers of a forward pass with an edit to a single layer outside of that range of layers. We summarize this hypothesis as follows: *Many layers could store a fact, and it happens that some do.*

If this hypothesis were true, it would be surprising because one cannot arbitrarily swap layers in a Transformer model without greatly damaging model performance [39]. That is, it should matter where information enters the residual stream, since later layers strongly depend on receiving the right incoming information from prior layers. We leave it to future work to further investigate this hypothesis.

What do our results imply about using model editing to validate localization claims? We interpret our results to suggest that Causal Tracing *answers a different question* than model editing does. That is, Causal Tracing answers a question about where factual information is carried in representations in a Transformer forward pass, and this question turns out to be a different question than the *editing* question of where is best to intervene in the Transformer in order to change the factual information it expresses. It seems critical, then, to carefully formalize the questions that one wishes to answer before (1) validating the results of localization via editing or (2) motivating the design of an editing method via localization, because the conclusions that can be drawn from a particular localization method might not be relevant for the performance of a given model editing method. This would not imply the conclusions from the localization analysis are invalid, though. For instance, we believe Causal Tracing reveals interesting insights about where MLP representations contain factual information (see Figs. 1 and 2). We only wish to suggest that localization analysis might answer a different question than the question answered by model editing.

These observations may have implications for the array of studies that validate their localization analysis by manipulating a certain model behavior via an intervention on the model component recommended by the analysis [29, 18, 2, 1, 26, 34, 8, 19, 7, 36, 4, 21]. Do model editing experiments provide *additional* evidence for claims about which model components are responsible for certain behaviors? If localization and editing answer different questions, editing experiments will not provide further evidence for localization conclusions.

[2]This tests if one model explains more of the variance than another model which has only a subset of the first's covariates (here, tracing effect and edit layer vs. only edit layer).

[3]Although, datapoint-level regression would provide stronger evidence that tracing effects predict which token representation is best to optimize with ROME (and rule out other confounders such as the edit layer).